

5(a)	any two from: 1 cyclic involves (only) PS1 and non-cyclic involves PS1 and PS2 ; 2 cyclic produces only ATP and non-cyclic produces ATP and reduced NADP ; 3 cyclic does not involve photolysis of water and non-cyclic does involve photolysis of water ; A photolysis only occurs in non-cyclic	2
5(b)(i)	as a, control / reference tube, for comparison ; R control variable	1
5(b)(ii)	<i>tube 2 changes to blue</i> 1 (plant carries out) photosynthesis ; 2 uses up CO₂ in the, Calvin cycle / light independent stage or less CO₂ causes pH to increase ; I oxygen <i>tube 3 changes to yellow</i> 3 (snail carries out) respiration ; 4 produces CO₂ in, the link reaction / Krebs cycle or more CO₂ causes pH to decrease ; A more CO₂ acidifies solution I oxygen <i>tube 4 stays green</i> 5 both photosynthesis and respiration or CO₂ produced in respiration is used in photosynthesis or oxygen produced in photosynthesis is used in respiration ;	5

Question	Answer	Marks
5b(iii)	any three from: 1 no / less, photosynthesis, so, no / less, CO₂ used ; 2 detail ; e.g. no light dependent stage / no non-cyclic photophosphorylation no / less, Calvin cycle / light independent stage / carbon (dioxide)fixation 3 respiration occurs producing CO₂ ; 4 respiration rate slower, in aquatic plants / test-tube 6, than, pond snails / test-tube 7 ; 5 (so) slight decrease in pH ;	3